Frontend & API contracts

Responsive globe navigation with explicit request boundaries and empty states.

PROPOSED • Not implemented • Scientific goal, scoring weights and hosting provider remain undecided

01

Frontend modules

GlobeView: global discovery and local Jezero view. MissionPanel: base, crew profile, limits and goal. LayersPanel: imagery, terrain and geology. StopPanel: tasks and evidence. FieldView: numbered stops, route and checklist.

02

State and accessibility

Walking astronauts are the primary workflow; crew vehicles and robotic support are optional profiles. Goal starts null. Keep map, mission and cached-data state separate. Support keyboard navigation, stale-request cancellation and explicit loading/error states.

03

Catalog endpoints — proposed

GET /v1/craters for bounded global search; GET /v1/craters/{id} for context; GET /v1/regions/jezero/stops for local science stops; GET /v1/layers for coverage and provenance. Support bbox filters and cursor pagination.

04

Analysis endpoints — proposed

POST /v1/rankings compares local stops using base, goal_id, crew/support profile and limits. Return component scores, planned science tasks, explanations and versions. Missing goal blocks ranking but not browsing. No global traversability claim.

05

Route jobs — later extension

POST /v1/routes creates a bounded analysis job and returns 202 + job_id. GET /v1/jobs/{id} reports progress; GET /v1/routes/{id} returns completed results. Define timeouts, cancellation, expiry and retry-safe submission IDs.

06

Browser performance

Cluster or tile markers at low zoom and fetch details on demand. Debounce searches, cache immutable assets and preserve attribution. Provide a list-based fallback when WebGL is unavailable. Do not download the full catalog into every browser.

Read technical-brief.md for linked sources, assumptions and competition timing.